Eddie Tang

609-480-1029 | eddietang2314@gmail.com | linkedin.com/in/ejtang | github.com/EDED2314 | Skillman, NJ

Education

University of Illinois Urbana-Champaign Urbana, IL
*BS in Electrical Engineering | GPA: **4.0** | James Scholar Honors* *Aug. 2025 – May 2028*
Minor in Computer Science & Semiconductor Engineering

Montgomery High School (MHS) Skillman, NJ
*High School Diploma | GPA: **3.92** | SAT: 1560 | 12 AP exams (all scores 5)* *Sept. 2021 - Jun. 2025*

Relevant Coursework

Analog Signal Processing (ECE210), Intro to Computing (ECE120), Differential Equations (MATH285), Multivariable Calculus (MATH214), Quantum Physics (PHYS214), Thermodynamics (PHYS213)

Projects

CAM-MK3 | *Illinois Space Society - Spaceshot Avionics | KiCAD, C++* Oct. 2025 – Present
- Third major revision of a compact live-video camera transmitter PCB (60x40mm) for rocket avionics; co-designed schematic, layout, and power architecture in KiCAD with a 3-person team.
- Assembled and debugged prototypes (hot-air reflow + hand solder), and verified electrical & signal integrity with bench tools (oscilloscope, power supplies, multimeter)
- Developed firmware drivers for TVP5151 analog video decoder and ESP32-P4 LCD_CAM interface; validated NTSC → digital pipeline producing 720×480 color frames (YUV422/YUV420).
- Integrated SW&HW H.264 encoding on ESP32-P4 and near-completed UVC USB video streaming for host.
- Implemented and validated cross-board I^2C communication and 433 MHz transceiver drivers; board designed to fit rocket avionics bay and scheduled for upcoming flight tests.

Other PCBs | *Illinois Space Society - Spaceshot Avionics* | KiCAD, C++, Python Oct. 2025 – Present
- Designed schematic, layout, and wiring of GATOR, an improved power distribution (for 24V) and ground-station PCB to track rocket position and adjust Yagi antenna direction with servos, improving tracker accuracy.

TARC Avionics V1, V2, V3 | EasyEDA, C++ Jun. 2022 – Jun. 2025
- Designed, assembled, and programmed custom PCB and attitude and heading reference system software, implementing 6DOF sensor fusion, for active flight control to variable target-apogees with 80% success rate.
- Developed and implemented quaternion-based algorithm to correctly integrate global acceleration for altitude.
- Coded Python software-in-the-loop-simulations to analyze and test prototype logic & algos using past flight data.
- Ranked top 50 nationally at TARC Nationals against 1000 teams. Qualified twice for Nationals.

Experience & Leadership

Electric Sensors Project Lead | *Illini Solar Car* Jan. 2026 – Present
- Led 8-member team and co-designing a Tire Pressure Monitoring System in KiCAD
- Researched implementation of low-power operations, system-design, and protocols for select sensors and chips.
- Defined validation KPIs and test plan; coordinated team schedule and deadlines.

Research Assistant (Paid), Mechanical Engineering Department | *Rowan University* Summer 2025
- Investigated effects of point defects in thermoelectric materials on lattice thermal conductivity using MD sims.
- Employed Green-Kubo method to calculate thermal conductivities from heat flux autocorrelation functions.
- Reduced simulation-to-data turnover times by creating Slurm workflow with Matplotlib, LAMMPS, and ASE.
- First author on in-preparation manuscript investigating point defect effects on thermal conductivity in Si and Ge.

Laboratory Learning Program, Mechanical & Aerospace Department | *Princeton University* Summer 2024
- Performed DFT calculations with VASP on WO_3 surfaces to advance plasma-assisted NH_3 synthesis research.
- Elucidated simulation data through custom Slurm workflow using ASE, Matplotlib, and NumPy in Python.
- Co-authored a peer-reviewed publication in ACS Energy Letters (Impact Factor: 19.3).

Vice President, Aerospace Club & Founder, Rocketry Team | *MHS* Sept. 2023 – Jun. 2025
- Led weekly meetings with 30+ students to engineer rockets; raised $15K+ through STEM fundraiser initiatives.
- Increased member attendance and participation by 10% by fostering a close-knit, community-first team culture

Skills

Programming Languages: Python, C++, C, Dart, Go, Bash, Java, JSON
Software: Linux, Git, KiCAD, EAGLE, OnShape, NX Siemens, Platform-IDE, MCUxpresso, Visual Studio Code, VASP, LAMMPS, Slurm, Jupter Notebook, Excel, Slack
License: HAM Radio Technician Class, Call sign: KE2CNQ
Fluent Languages: English, Chinese